Hydrological Processes

WILEY

RESEARCH ARTICLE OPEN ACCESS

Comparing Seasonal Soil Water Storage and Flow Processes Under Different Soil Conditions

Amila Ljutic[1] | Genevieve Ali[1,2,3] | Laura Van Eerd[4] | Merrin Macrae[5] | Claudia Wagner-Riddle[1]

[1]School of Environmental Sciences, University of Guelph, Guelph, Ontario, Canada | [2]Department of Earth and Planetary Sciences, McGill University, Montreal, Quebec, Canada | [3]Department of Geography, McGill University, Montreal, Quebec, Canada | [4]School of Environmental Sciences, University of Guelph, Ridgetown, Ontario, Canada | [5]Department of Geography and Management, University of Waterloo, Waterloo, Canada

Correspondence: Amila Ljutic (aljutic@uoguelph.ca)

Received: 11 March 2025 | **Revised:** 16 January 2026 | **Accepted:** 21 March 2026

Keywords: agriculture | cover crop | matrix flow | preferential flow | soil compaction | soil water storage

ABSTRACT

Soil moisture dynamics are known to vary greatly in space and time, especially given soil–plant-water and human interactions in agricultural environments. Knowledge gaps remain surrounding how soil moisture processes change under different soil conditions and land management practices. In this study, soil moisture storage and flow were examined over seasonal timescales in 2021 and 2022 in Southern Ontario, Canada under three soil agricultural management treatments—control (no-till), cover cropped and compacted soil. To compare soil moisture patterns, experimental plots under each treatment (three replicates) were characterized based on vertical soil moisture distributions in the root zone (0–60 cm). Monotonic versus non-monotonic soil moisture profiles were taken as an indication of different matrix and lateral flow pathways when field capacity was exceeded. Results showed that despite soil water storage trends being similar inter-treatment (i.e., similar 'average' changes), lower water storage capacity was measured on compacted plots. Soil moisture pattern analysis revealed 95% of the study period was characterized by non-monotonic soil moisture profiles. When monotonic patterns were identified, the control and cover crop treatments were more often associated with soil moisture that increased monotonically with depth (4% of time), attributed to better water infiltration and storage, whereas the opposite was observed for the compacted treatment. Although no dominant flow pathway could be identified for 75% of timesteps, flow pathways were more active in the spring and fall seasons, as expected, with high spatial variability of inferred instances of matrix and lateral preferential flow. Both years considered, the control and cover crop treatments each had four times more observations of capillary rise compared to the compacted treatment. The analysis of vertical soil moisture patterns therefore revealed process insights that may otherwise be overlooked when solely focusing on total soil water storage or water budget analyses.

1 | Introduction

The factors that drive water infiltration into soil are widely studied, as are subsequent fluctuations in soil moisture storage and flow generation processes that can occur within the vadose zone (e.g., Seneviratne et al. 2010; Arora et al. 2019). The strong variability of soil moisture in space and time continuously motivates additional study, given the implications for runoff generation (Dunne et al. 1975), soil erosion (Moragoda et al. 2022), drought susceptibility (Ruosteenoja et al. 2018), nutrient cycling (Bauke et al. 2022) and contaminant persistence and export (Lewan et al. 2009), among other processes (Seneviratne et al. 2010). A better understanding of soil water dynamics is especially important in agricultural

Abbreviations: CC, cover crop; CP, compacted; CT, control; FC, field capacity; FP, flow pathway; MP, moisture pattern; SM, soil moisture.

systems, as the productivity of most crops around the world depends on soil water in the vadose zone (Rosegrant et al. 2009). Achieving a better understanding, however, requires that soil moisture data be used not only to quantify soil water storage, but also to draw inferences on shallow subsurface flow processes and the influence of agricultural practices on those processes.

Shallow subsurface water flow is generally characterized as being either matrix flow or preferential flow, with both involving the movement of soil water in the vertical or lateral direction. Following infiltration, a wetting front that propagates uniformly through soil micropores is referred to as matrix flow (Allaire et al. 2009). Preferential flow displaces water at a significantly faster rate than matrix flow, as water bypasses soil micropores and flows through a small fraction of the soil profile (Stumpp and Kammerer 2022). Vertical preferential flow occurs via cracks, fissures or vertically oriented macropores in the soil, which are typically continuous soil pores of larger diameter (> 75 μm) or cavities created by earthworm burrows or root channels (Beven and Germann 1982; Gjettermann et al. 1997; Pituello et al. 2016). As for lateral preferential flow, it is typically the transport of water through horizontally-oriented macropores or along textural discontinuities in the soil, such as the interface between two soil horizons (Shein et al. 2009; Zhu et al. 2014). Differentiating between matrix and preferential flow is especially important in nutrient-rich (or more generally, contaminant-rich) environments, such as agricultural landscapes, because different soil water flow processes are associated with different nutrient and contaminant export potentials (Clothier et al. 2008). While subsurface flow processes are difficult to measure directly in the field, some can be inferred—that is, their presence or absence can be ascertained—from soil moisture data.

Since field capacity is the amount of water that soil can hold after excess (gravitational) water has drained out, some studies have identified the presence of subsurface matrix flow when this critical soil moisture threshold is exceeded (Kampf 2011; Lai et al. 2018; H. Zhang et al. 2025). Timeseries of multi-depth soil moisture can be used to examine vertical variations in water content and identify where soil moisture is highest in the soil profile. For each timestep, soil moisture can be described as increasing or decreasing monotonically with depth (Peterson et al. 2019; Rosenbaum et al. 2012). Pattern variations can then be linked to different process hypotheses, such as top-down wetting front migration in the soil profile or bottom-up wetting regimes controlled by groundwater and capillary rise, all of which are manifestations of vertical matrix flow (Lu and Likos 2004). The presence of lateral preferential flow can be inferred by monitoring the sequence of soil moisture response, particularly when non-monotonic patterns are observed (Graham and Lin 2012; Lin and Zhou 2008). Although these strategies can provide insights on soil water flow processes, they have not yet been used in combination in a single study, especially not in agricultural environments.

Agricultural practices—such as tillage, inorganic and organic fertilizer application, cover cropping and soil conditions such as compaction—are known to alter soil characteristics and hydrological functioning (Ali et al. 2018; Nagy et al. 2018). Soil compaction is the physical degradation of soil structure, which reduces air and water infiltration (Nawaz et al. 2013). Although soil compaction can occur naturally, it is especially common and potentially severe in agricultural landscapes due to repeated passes of heavy agricultural machinery (Keller et al. 2019). Results from previous studies show that compaction can increase soil bulk density, changing the way that soil water behaves and the ability of plant roots to propagate (Blanchy et al. 2020; Colombi et al. 2018). However, knowledge regarding the impacts of soil compaction on seasonal soil water storage dynamics and dominant subsurface flow processes is limited.

Cover cropping involves planting crops, often broadleaves, grasses or legumes, for the services they offer rather than for harvest (Van Eerd et al. 2023). Long-term cover crop use has been shown to increase soil water holding capacity as cover crop biomass (i.e., organic matter—leaf litter and root exudates) accumulates on the soil surface and increases soil organic matter content over time (A. Basche and DeLonge 2017; Chalise et al. 2018). This residue left on the soil surface can also act as a barrier; effective at reducing water loss due to evaporation (Alonso-Ayuso et al. 2014). Tribouillois et al. (2022) found that short-term and long-term cover crops were able to decrease drainage in the fall season by 8.7% and 9%, respectively, due to increased evapotranspiration, suggesting that cover cropping can potentially limit the amount of nutrient leaching. However, cover cropping has also been shown to alter soil physical properties such as increasing porosity (Blanco-Canqui and Ruis 2020) and aggregate stability (Rieke et al. 2022), while decreasing bulk density (Chalise et al. 2018), all of which can increase soil water storage and matrix flow (A. D. Basche and DeLonge 2019). Furthermore, several studies have shown that cover crops can increase macroporosity of the soil by promoting earthworm activity and the development of root channels (Blanco-Canqui et al. 2011; Haruna et al. 2022; Ogilvie et al. 2021), which coincides with an increase in the amount of preferential flow (Y. Zhang et al. 2018). Although preferential flow can be beneficial, it may also promote increased nutrient and contaminant downward transfer effectively bypassing the root zone (Clothier et al. 2008). These conflicting findings justify the need for further analysis on how cover crops influence soil water storage and flow processes in contrasting seasons.

The goal of this study was to characterize seasonal soil water storage and infer subsurface water flow processes by using high-frequency, multi-depth soil moisture data collected on experimental farmland, with and without cover crops and compaction treatments. Specifically, the following research questions were addressed:

1. How does soil water storage vary across seasons within a treatment and between treatments?
2. What is the seasonal prevalence of uniform and non-uniform (i.e., preferential) lateral subsurface flow processes in the vadose zone?
3. To what extent, if any, are soil water storage dynamics and dominant subsurface flow processes impacted by the presence/absence of cover crops and compaction? Does within-treatment variability exceed between-treatment variability?

2 | Materials and Methods

2.1 | Study Area Description

The study was conducted at a long-term micrometeorological site (43°38′20.2″N 80°24′36.9″W) at the Ontario Crops Research Centre (Field E26) near Elora, Ontario, Canada (Gao et al. 2023) (Figure 1a). The region is classified through the Köppen climate classification system as a warm summer humid continental climate (Dfb) (Kottek et al. 2006). The field site is level (< 0.5% slope) and has an elevation of approximately 376 m above sea level (Saso et al. 2012). The area experiences an average annual air temperature of 6.7°C with an average precipitation of 946 mm per year, 148 mm as snow (S. Brown et al. 2021). The average annual daily air temperatures were 8.0°C (2021) and 6.9°C (2022), with growing season (May–October) temperatures of 16.4°C (2021) and 15.7°C (2022) and non-growing season temperatures of −0.2°C (November 2020–April 2021) and −1.9°C (November 2021–April 2022) (Environment Canada 2023; OMAFA 2022). Both years were drier than 30-year long term averages, with total precipitations of 730 and 580 mm in 2021 and 2022, respectively (Environment Canada 2023).

The soil is characterized as Grey-Brown Luvisol based on the Canadian Soil Classification System (Soil Classification Working Group 1998) or Albic Luvisol (IUSS Working Group W.R.B 2006) and surface soil (0 to 30 cm) at the experimental field is an imperfectly drained Guelph silt loam (S. Brown et al. 2021). Additional soil characteristics can be found in Table 1. The field has been tile drained since the 1960s: the tiles are located at approximately 80 cm depth, about 15 m from each other (Pawlick et al. 2019). The site is on a three-year diverse crop rotation of corn, soybean and winter wheat, with cover crops planted, interseeded with corn and after winter wheat that started in 2018.

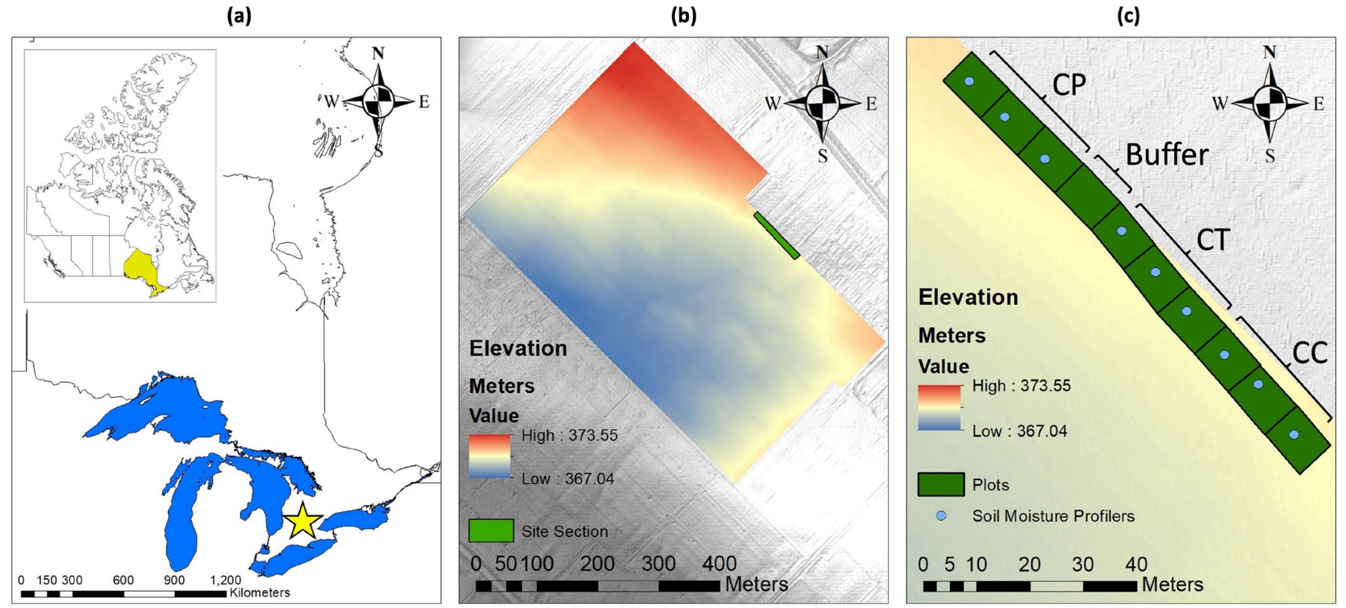

FIGURE 1 | (a) Map of Ontario (ESRI Canada, Natural Earth Vector. 'Provinces and Territories of Canada' [data layer]. 1:2000000. August 11, 2022. URL (Retrieved: August 16, 2022)) with yellow star to indicate the location of the research centre (Elora, Ontario) and Great Lakes shown in blue (sparksa_umich. 'great_lakes' [data layer]. 1:24000. April 15, 2019. URL (Retrieved: August 16, 2022)). Inset map of Canada has the province of Ontario highlighted in yellow. (b) Elevation map of the experimental field with location of study plots indicated as a green rectangle. (c) Enlarged section of elevation map with individual plots and soil moisture profilers shown. Treatments labelled as CP (compaction), CT (control) and CC (cover crop). All panel maps were created using ArcMap (version 10.6).

TABLE 1 | Soil characteristics at the long-term micrometeorology site at the Ontario Crops Research Centre adapted from S. Brown et al. (2021).

Mineral soil horizons	Horizon depth (cm)	Range of horizon thickness (cm)	Textural class	%Sand	%Silt	%Clay	Saturated hydraulic conductivity $cm\,h^{-1}$
Ap	0–32	20–34	Silt loam	38.0	54.5	7.5	3.1 ± 2.2
Bt	32–61	20–30	Loam	44.7	40.3	15.0	0.96 ± 0.5
IICk	61+	—	Loam	49.4	38.1	12.5	2.6 ± 0.96

Note: The average of three soil core samples were used to characterize the textural class and measure saturated hydraulic conductivity. Errors (±) are a single standard deviation from the mean. Detailed soil analysis methods used to derive this data are in S. Brown et al. (2021).

2.2 | Experimental Setup

Nine neighbouring plots (8 m × 10 m each) were delineated at the edge of field E26 (Figure 1c). Three plots had a cover crop planted (CC-1, CC-2 and CC-3), the middle three plots were control plots (CT-1, CT-2, CT-3) and the final three plots located in the northwest portion were mechanically compacted on two separate occasions (CP-1, CP-2 and CP-3). The CC, CT and CP were considered as independent soil treatments. For the CP plots, the first round of mechanical compaction was completed in October 2020 with two passes of a 1180 kg steel water-filled roller. The second round of compaction was performed in March 2021 (a few days after soil moisture sensor installation), with two passes of a gas plate compactor. Both tools are used in construction, for example, to compact aggregates prior to pavement. A buffer zone (8 m × 16.9 m) was set up between the control plots and the compacted plots, to prevent the control plots from being affected by the compaction equipment. In late August 2020, following harvest of winter wheat, a 4-way cover crop was disked into the soil on all plots: it contained 4 species: crimson clover (*Trifolium incarnatum* L.), cereal rye (*Secale cereale* L.), daikon radish (*Raphanus sativus* L. *var. longipinnatus*) and oat (*Avena sativa* L.). This cover crop was chemically terminated in all nine plots with glyphosate and dicamba on May 10th, 2021. For the specific cover crop management, please refer to Menheere (2023). Soil was not disturbed other than during planting. During the first year of study, in 2021, soil moisture data collection (see details below) took place between March 22nd and November 30th, effectively spanning the entire 2021 growing season. A grain corn (*Zea mays* L.) cash crop was planted on May 18th, 2022, in rows 76 cm apart. To avoid damaging the sensors, corn was hand-planted: first, a garden hoe was used to dig 5 cm deep and then, a walk-behind seeder (Cole Planet Jr. Push Seeder, Albany GA) was used to administer the corn seed. A liquid starter fertilizer (Alpine G241-S with N:P:K:S 8–28–5–1) was applied at 14.926 kg/ha on May 19th, 2021 and Urea Ammonium Nitrate (175 kg N/ha) was applied on June 18th, 2021. Once corn had reached the V6 leaf stage, a 2-way cover crop consisting of crimson clover and annual ryegrass (*Lolium multiflorum* L.) was broadcast seeded at 6.7 and 33.6 kg/ha, respectively, exclusively on the CC plots. Corn was harvested on November 5th, 2021 and the following spring on May 10th, 2022, the cover crop was chemically terminated with glyphosate. In the second year, in 2022, soil moisture data collection took place between March 18th and October 15th. A soybean (*Glycine max*. (L.) Merr.) cash crop was planted on May 24th, 2022 in rows approximately 20 cm apart using the same planting procedure as corn. Soybean was harvested on October 13th, 2022 and winter wheat was planted afterwards. This field has been under different management practices over more than five decades, with corn being the dominant crop in the last 20 years. Soybean and winter wheat have also been grown in rotation with corn during some years. Tillage has included no-till (year 2000 to 2010) and disking (5–10 cm deep) after corn, with soybean and winter wheat typically planted with no-till since then. In the years preceding this experiment, corn was grown in 2018 with disking applied before soybean planting in 2019, no-till winter wheat was grown over fall 2019 to summer 2020. The study plots were not tilled at any point during the experiment, except for the garden hoe used to 5 cm depth. Detailed historical site management can be found in earlier publications (Machado et al. 2021; Ranathunga et al. 2025).

High-frequency soil moisture data was collected using a multi-profile soil moisture instrument (Dataflow Systems Ltd) installed in the middle of each plot in March 2021. The instrument is composed of a sensing rod attached to a data logger. Along the sensing rod are four discrete sensors that can measure soil moisture at four target depths: the 20 cm and 30 cm sensors were located within the Ap horizon, while the Bt horizon contained the 40 and 60 cm sensors. Recording frequency was set to 15 min during the study period. During weekly field visits, data were retrieved from the loggers via wireless Bluetooth connection. It should be noted that the sensing rods are deployed into fibreglass access tubes that act as protective casing: narrow holes (diameter~2 cm) therefore had to be drilled in the soil to a depth of 60 cm using a rotary hammer with a 0.75-in. carbide bit. The sensor's factory settings were used to avoid the disruptive soil sampling which would have been required at several different points on each plot for soil-specific calibration.

In addition, an irrigation experiment was conducted on October 20, 2022 (outside of the data collection period—ended October 13, 2022) to measure depth-specific field capacity on our experimental plots. A detailed procedure and a table with all field capacity values (Table S1) can be found in Supporting Information S1.

2.3 | Analyses Pertaining to Specific Research Questions

To examine temporal variations in soil water dynamics, seasons defined in relation to crop stages were used: prior to planting (early spring: March 20th to May 5th, 2021), planting and early growth/emergence (late spring: May 6th to June 20th), crop growth and establishment (early summer: June 21st to August 6th), full canopy closure (late summer: August 7th to September 22nd), harvest (early fall: September 23rd to November 6th) and post-harvest (late fall: November 7th to December 21st). As data collection in 2021 began on March 22nd and ended on November 30th, the early spring season is missing 2 days (March 20th–21st) and the late fall is missing 21 days of data (December 1st–21st). Data collection in 2022 began on March 18th and ended on October 15th, meaning the early fall is missing 20 days (October 16th to November 6th) and no data are available for late fall.

To specifically address research question 1 (i.e., how soil water storage varies across seasons), total soil water depth was calculated for each logged 15-min timestep in 2021 ($n = 24\,266$) and 2022 ($n = 20\,246$) to provide an estimate of how much water was stored in the top 60 cm of the soil profile. Total soil water depth is given by the formula:

$$\text{Total soil water depth (cm)} = \sum_{i=1}^{N} \left(S_i \times T_i\right)$$

where the index (i) represents a soil layer (out of a total of $N = 4$ layers) associated with a soil moisture sensor. The thickness (T) of each layer (i) is the distance between two consecutive sensors: the 20 cm sensor represents a 20 cm-thick soil layer between 0 and 20 cm; the 30 cm and 40 cm sensors represent 10 cm-thick layers between 20–30 and 30–40 cm, respectively and the 60 cm represents a 20 cm-thick layer between 40 and 60 cm. Soil moisture (S) is measured by each sensor and expressed in percent volume (%vol) units which were converted to $m^3 m^{-3}$. For each experimental plot, a timeseries of total soil water depth (expressed in cm) was created to illustrate fluctuations in soil water storage over time. Descriptive statistics of total soil water depth, including the minimum, maximum, mean, median, standard deviation and coefficient of variation in each season, were computed and compared. All computations were performed in MATLAB (version R2021a).

To address research question 2 (i.e., seasonal prevalence of uniform and non-uniform (i.e., preferential) lateral subsurface flow processes in the vadose zone), the variability of soil moisture with depth was examined at each individual time step. Soil moisture profiles were classified in two separate ways: based on their moisture pattern (MP classification) of variability and based on their inferred dominant water flow pathway (FP classification) (Figure 2). The MP classification was done in line with previously published studies (e.g., Graham and Lin 2012), which labelled soil moisture profiles as either monotonic (in-sequence) or non-monotonic (out-of-sequence) with depth. Differences of 2.5% (approximately one twentieth of the soil moisture range) or less in soil moisture between consecutive depths were not considered significant and therefore used to label soil moisture profiles as uniform (class MP-3). Conversely, greater differences were used to label non-uniform soil moisture profiles (classes MP-1, MP-2 and MP-4, see Figure 2a). Maintaining the 2.5% threshold across depths was a methodological decision for consistency and does not assume that similar dominant processes occur at each depth. The FP classification focused on the depth(s) at which soil moisture was highest and whether soil moisture at that depth exceeded field capacity, which could lead to the inference of different subsurface flow pathways (Figure 2b). Wetting front propagation (FP-1), which is uniform downward percolation from the soil surface via soil micropores (Allaire et al. 2009), was inferred when soil moisture was the highest at 20 cm depth and exceeded field capacity. Lateral

(a) MP Classification

Classes	Description	Profile Example
MP-1	Soil moisture increases monotonically with depth	
MP-2	Soil moisture decreases monotonically with depth	
MP-3	Soil moisture is uniform (depth-to-depth difference ≤2.5%)	
MP-4	Non-monotonic	

(b) FP Classification

Classes	Description	Profile Example
FP-1	Wetting front propagation: soil moisture is highest at 20 cm; and SM > FC at 20 cm depth	
FP-2	Lateral preferential flow to 30 cm: soil moisture is highest at 30 cm; and SM > FC at 30 cm	
FP-3	Lateral preferential flow to 40 cm: soil moisture is highest at 40 cm; and SM > FC at 40 cm	
FP-4	Capillary rise: soil moisture is highest at 60 cm; and SM > FC at 60 cm	
FP-5	Wetting front propagation and capillary rise: highest and second highest soil moisture at 20 or 60 cm; SM > FC at both depths	
FP-6	No dominant flow pathway: FC not exceeded at depth with highest soil moisture (SM < FC)	

FIGURE 2 | Summary of soil moisture profile classifications performed in this study. (a) Soil moisture pattern (MP) classification and (b) flow pathway (FP) classification. The third column includes one or two profile examples, where the darkest blue represents the depth with the highest soil moisture content. The water droplet in the FP classification represents soil moisture (SM) at the corresponding depth exceeding field capacity (FC).

preferential flow at 30 cm (FP-2) and 40 cm (FP-3) depth was inferred when soil moisture was highest at a middle depth (i.e., the middle depth was wetter than soil immediately above or below) and exceeded field capacity. Lateral preferential flow at 20 cm or 60 cm could not be inferred, as we did not have information about soil water content above 20 cm or below 60 cm and therefore could not make inferences about bypass flow at those depths. Capillary rise (FP-4), which is the upward movement of free water to unsaturated finer pore spaces driven by soil water potential, was inferred when soil moisture was the highest at 60 cm depth and exceeded field capacity. Custom MATLAB scripts (version R2021a) incorporating the criteria listed in Figure 2 were written to automate the soil moisture profile classification procedures. Categorical timeseries were created to illustrate the temporal variability in the presence/absence of the different MP classes and FP classes. This procedure was repeated for all plots.

To address research question 3 (i.e., to explore the extent to which soil water storage dynamics and dominant subsurface flow processes are impacted by the presence/absence of cover crops and compaction), the results of the aforementioned analyses were compared within and between treatments. Two sub-questions were examined, specifically: does within-treatment variability exceed between-treatment variability, with respect to (i) soil water storage and the (ii) frequency of occurrence of different MP classes or FP classes?

To address sub-question (i), each treatment was considered independently for both years. For total soil water depth (calculated for each 15-min timestep), Spearman's rank correlation coefficients (r_S) were calculated in MATLAB to determine the extent to which replicates of the same treatment behaved similarly. Spearman's rank correlation is a non-parametric test that measures the strength and direction (positive/negative) of the relationship between two ranked variables (Triola et al. 2018). In the present study, a correlation coefficient between total soil water storage of two replicates of the same treatment was deemed statistically significant when p-values were less than 0.05. Based on the computed values of r_S, the similarity of replicates—in terms of their soil water storage dynamics—was qualified as perfect ($r_S = 1$), strong ($0.7 \le r_S < 1$), moderate ($0.4 \le r_S < 0.7$), weak ($0.1 \le r_S < 0.4$) or nonexistent ($r_S \le 0.1$). To assess between-treatment variability, replicate data was first averaged for each treatment. Spearman's rank correlation coefficients were then calculated between the treatments to determine the degree to which they behaved similarly.

To address sub-question (ii), the Pearson's chi-square test of homogeneity was conducted in SPSS (version 28) to assess whether replicates of the same treatment had similar MP and FP class frequency counts in each year. The chi-square test of homogeneity tests the claim that different populations have the same proportions of some characteristics (Triola et al. 2018). Data for a chi-square test must be counts (not percentages) and set up in a contingency table, which is a table comprised of frequency counts of categorical data corresponding to two different variables. Assumptions for the Pearson's chi-square test of homogeneity are that samples are randomly selected from different populations, observations/counts are independent, categorical variables are mutually exclusive and there is sufficient data to ensure that expected cell frequency (in the contingency table) is at least 5 observations per cell (Triola et al. 2018). In the present study, when the expected cell frequency condition was violated, the Fisher Exact Test was conducted (Triola et al. 2018). It should be noted that the chi-square test is sensitive to large numbers of data points and may lean towards statistical significance if sample sizes are too large (Frieman et al. 2017). Hence, the data for the 2 years encompassed by this study (2021 and 2022) were tested separately. To examine within-treatment variability, the null hypothesis was that replicates of the same treatment had the same frequency counts of pattern or flow occurrences. The alternative hypothesis was that replicates of the same treatment had different frequency counts of pattern or flow occurrences. The null hypothesis was rejected when the test p-value was less than 0.05. To examine the variability (or lack thereof) of pattern or flow occurrence between treatments (i.e., between-treatment variability), the frequency counts of all replicates of the same treatment were aggregated. For example, for the CT treatment, the frequency counts of each MP class for plots CT-1, CT-2 and CT-3 were summated, to obtain an aggregated value of each MP class for the CT treatment. The Pearson's chi-square test of homogeneity was then used to test for statistically significant differences in patterns and inferred flow processes between treatments.

3 | Results

3.1 | Soil Water Storage Variations

Total soil water depth timeseries, which include a seasonal breakdown for both years, are displayed in Figure 3. In 2021, two large spikes in the early spring season are apparent, while other water storage peaks throughout the late spring to late fall are not as prominent. All plots experienced a plateau in water storage during the month of May, while a decline in water storage was evident on all plots throughout the month of August. Some soil water storage recharge beginning in the early fall season can also be observed. The cover crop (CC) treatment had higher fluctuations in soil water storage, as shown by the more prominent peaks in the associated timeseries (Figure 3c). In comparison, the compacted (CP) replicates had the least amount of changes (smaller peaks) in soil water storage between the months (Figure 3e). Similar general observations regarding inter-treatment differences were made in 2022 (Figure 3d,f), although some subtle differences across seasons were observed: more storage fluctuations were observed in late spring 2022 compared to 2021 and storage continued to decline into early fall 2022.

Summary statistics of total soil water depth for all plots are shown in Table 2. The overall maximum total soil water depth was observed in plot CC-1 in both 2021 and 2022, as well as the highest standard deviation and coefficient of variation. The overall minimum total soil water depth was measured on another cover crop plot (CC-3) for both years. In 2022, the CC-3 signal was an outlier caused by soil moisture recorded at the 40 cm depth which averaged ~10%vol: this influenced the total soil water depth of that plot. The CC treatment replicates had the highest coefficients of variation (i.e., variability in total soil

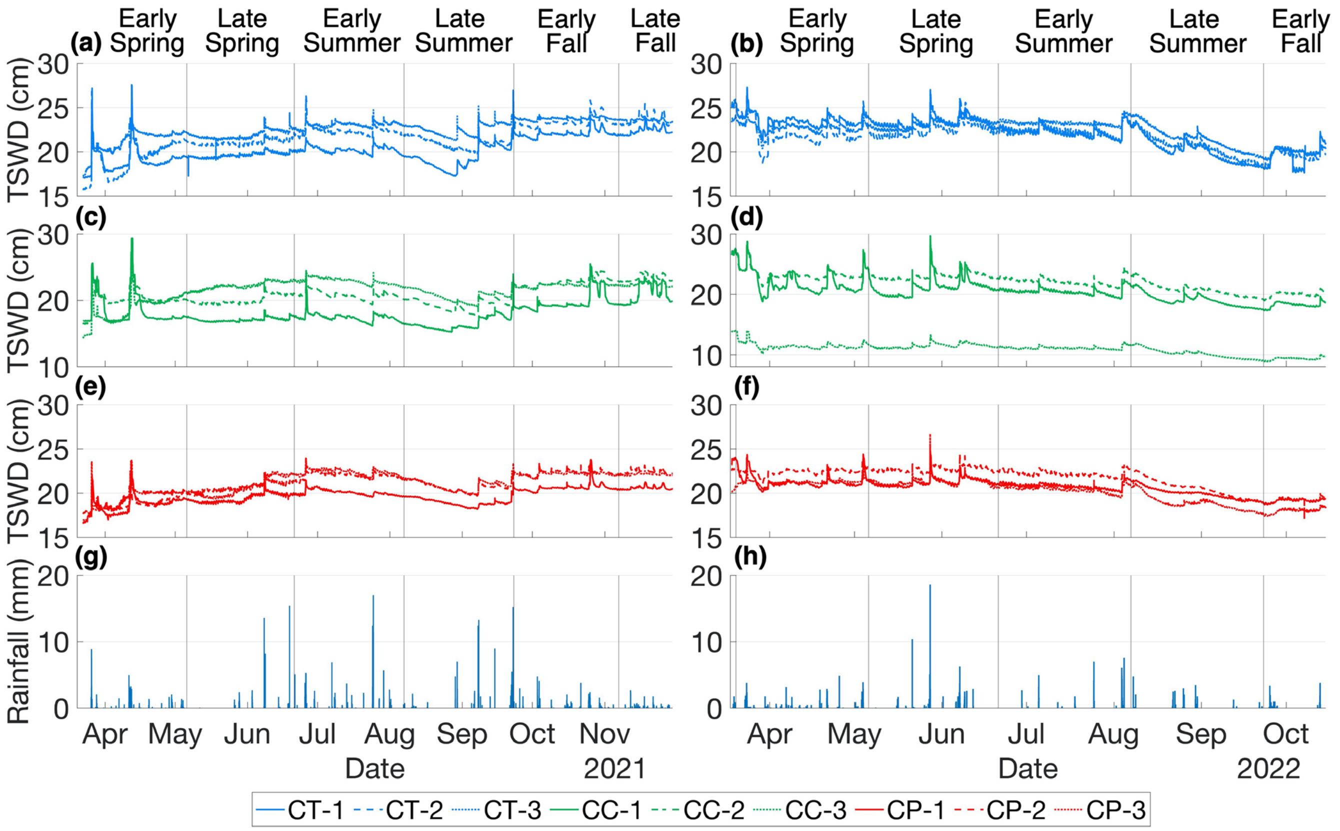

FIGURE 3 | Timeseries of total soil water depth as an estimate of how much water was stored in the top 60 cm of the soil profile for each plot, with seasonal breakdown. The columns correspond to 2021 (a, c, e) and 2022 (b, d, f) data, respectively. Panels in blue (a, b), green (c, d) and red (e, f) correspond to the control (CT), cover crop (CC) and soil compaction (CP) treatments, respectively, with each line showing one of three replicates. The last row of panels (g, h) show hourly rainfall data.

water depth) out of all treatment replicates. Plot CT-3 had the highest mean and median values. For both years, the CP treatment replicates had coefficients of variation that were the most similar (within 0.01) to one another compared to other treatments. Overall, the range of values across all plots and both years for the total soil water depth was 8.9–29.7 cm.

3.2 | Soil Moisture Profile Classifications

Results of the moisture profile pattern of variability (MP) classification for all plots and for each year (2021 and 2022) are shown as categorical timeseries in Figure 4a. For all plots, the temporally predominant class in 2021 and 2022 was the non-uniform moisture profile (i.e., MP-4, 95% of all timesteps). Conversely, the class describing 'uniform' soil moisture (MP-3) occurred most rarely (0.14% of all timesteps) and only on control and cover crop plots. In early spring, early and late summer of 2021, some experimental plots were characterized by periods with soil moisture increasing (MP-1) or decreasing (MP-2) monotonically with depth (Figure 4a). Notably, plot CC-1 was associated with the MP-1 class for most of the late summer of 2021. Although 2022 was also mainly characterized by the non-uniform moisture pattern (MP-4, 93%), the class corresponding to soil moisture decreasing monotonically with depth (MP-2) was more common (5%), compared to 2021 (0.5%). The MP-2 pattern was mainly observed on compacted plots in the early and late spring seasons, as well as during the transition from early to late summer. As for the MP-1 category (i.e., soil moisture increasing monotonically with depth), it was observed for a few non-compacted plots in the late summer and early fall of 2022. Overall, the control and cover crop plots had more MP-1 identified (4% of all timesteps for both treatments) compared to the compacted plots, which had higher MP-2 (6% of all timesteps).

Results of the flow pathways (FP) classification are displayed as categorical timeseries in Figure 4b. In early spring 2021, all cover crop (CC) plots and the CT-1 plot behaved very similarly. Otherwise, there was a lot of variability across all plots in 2021, with stark differences in some seasons. Notably, CP-1 had the most lateral preferential flow inferred at the 40 cm depth (FP-3, 54% of 2021), almost continuously from the beginning of early spring to the beginning of the late summer season. The CC-3 and CP-3 plots had simultaneous occurrences of the wetting front pathway (FP-1) for most of the early summer season. The wetting front pathway was common for all treatments in the early fall season, especially the cover crop treatment. Overall, 2022 had fewer occurrences of wetting front propagation (FP-1), lateral preferential flow at 40 cm (FP-3) and capillary rise (FP-4) flow patterns, compared to 2021 (Figure 4b). The late spring of 2022 was more active compared to the same season in 2021, with various flow pathways identified. Although the overall occurrence of lateral preferential flow was minimal, there was more at 30 cm (FP-2) in 2022, rather than at the 40 cm depth as

TABLE 2 | Descriptive statistics for total soil water depth (computed for every 15-min timestep) during the 2021 (March 22nd to November 30th) and 2022 (March 18th to October 15th) study periods.

Plot	Year	Min.	Max.	Mean	Median	SD	CV
CT-1	2021	17.04	27.56	20.28	20.05	1.48	0.07
	2022	17.66	27.28	22.07	22.57	1.81	0.08
CT-2	2021	15.67	26.86	21.63	21.66	1.72	0.08
	2022	18.2	27.03	21.64	21.84	1.48	0.07
CT-3	2021	17.22	25.16	22.61	22.88	1.1	0.05
	2022	19.03	26.16	22.38	22.87	1.29	0.06
CC-1	2021	15.23	29.34	18.12	17.43	1.96	0.11
	2022	17.35	29.68	20.46	20.38	1.88	0.09
CC-2	2021	16.45	27.38	20.9	20.61	1.67	0.08
	2022	18.97	28.06	22.15	22.42	1.53	0.07
CC-3	2021	14.29	27.34	21.5	22.12	1.75	0.08
	2022	8.91	13.98	10.92	11.11	0.91	0.08
CP-1	2021	16.56	23.79	19.65	19.78	0.93	0.05
	2022	18.71	25.25	20.68	20.86	0.93	0.04
CP-2	2021	17.66	23.44	21.06	21.37	1.34	0.06
	2022	17.01	24.55	21.7	22.26	1.29	0.06
CP-3	2021	16.92	23.97	21.2	21.71	1.29	0.06
	2022	17.42	26.69	20.22	20.68	1.29	0.06

Note: Descriptive statistics include minimum (Min.), maximum (Max.), mean, median, standard deviation (SD) and coefficient of variation (CV). All statistics, except CV, are expressed in cm.

was the case in 2021 (FP-3). The compacted plots generated the most wetting front propagation (FP-1, 14.3% of all CP timesteps) in 2022, especially in the spring. The early fall had less flow pathway activity in 2022, compared to 2021. In both years, the control and cover crop treatments each had a four times higher occurrence rate of capillary rise, compared to the compacted treatment (Figure 4b). A single dominant flow pathway in space (i.e., across all nine plots) and time (i.e., across the whole study period) was not identifiable. Rather, the 'no dominant flow pathway' category (FP-6, 75% of all timesteps) was predominant for all treatments, not only in 2021 but also in 2022. When dominant flow pathways were identified, there was high variability; however, the control and cover treatments had more identifiable process similarities compared to the compacted treatment.

3.3 | Within-Treatment and Inter-Treatment Variability

For both years, all treatments had strongly similar replicates in terms of their total soil water depth fluctuations, as shown by positive and strong correlation coefficients ($0.7 \leq r_S < 1$; Table S2). One exception is the cover crop (CC) treatment, which had the least similar replicates in 2021 ($r_S = 0.38$; Table S2). For replicates of the same treatment, frequency counts of soil moisture pattern class (MP) and flow pathway class (FP) occurrence were all significantly different (p-value < 0.05; Table S3), except for MP occurrence of CC-2 and CC-3. It should be noted that in this study, there were a few instances where expected frequency counts for a category were below 5, notably for the occurrence of some MP and FP classes for some replicate pairs: these instances are flagged with an asterisk in Table S3.

Correlation coefficients (p-value < 0.05) for treatment-averaged values of total soil water depth demonstrated strong, positive inter-treatment relationships in both years (Table 3). In each year, relative proportions of frequency counts of the soil moisture pattern classes and flow pathway classes were all significantly different between the treatments (Figure 5).

4 | Discussion

4.1 | Variations in Soil Water Storage: Seasonal and Soil Condition Influences

Fluctuations in total soil water depth during different seasons of the year were evident (Figure 3). This result was not surprising, as soil water storage is driven by the balance or imbalance between precipitation, drainage and evapotranspiration (Seneviratne et al. 2010). The plateau observed in the total soil water storage timeseries in May 2021 indicated that water outputs (i.e., evapotranspiration, gravitational drainage) from the root zone did not exceed water inputs to the root zone (i.e., precipitation),

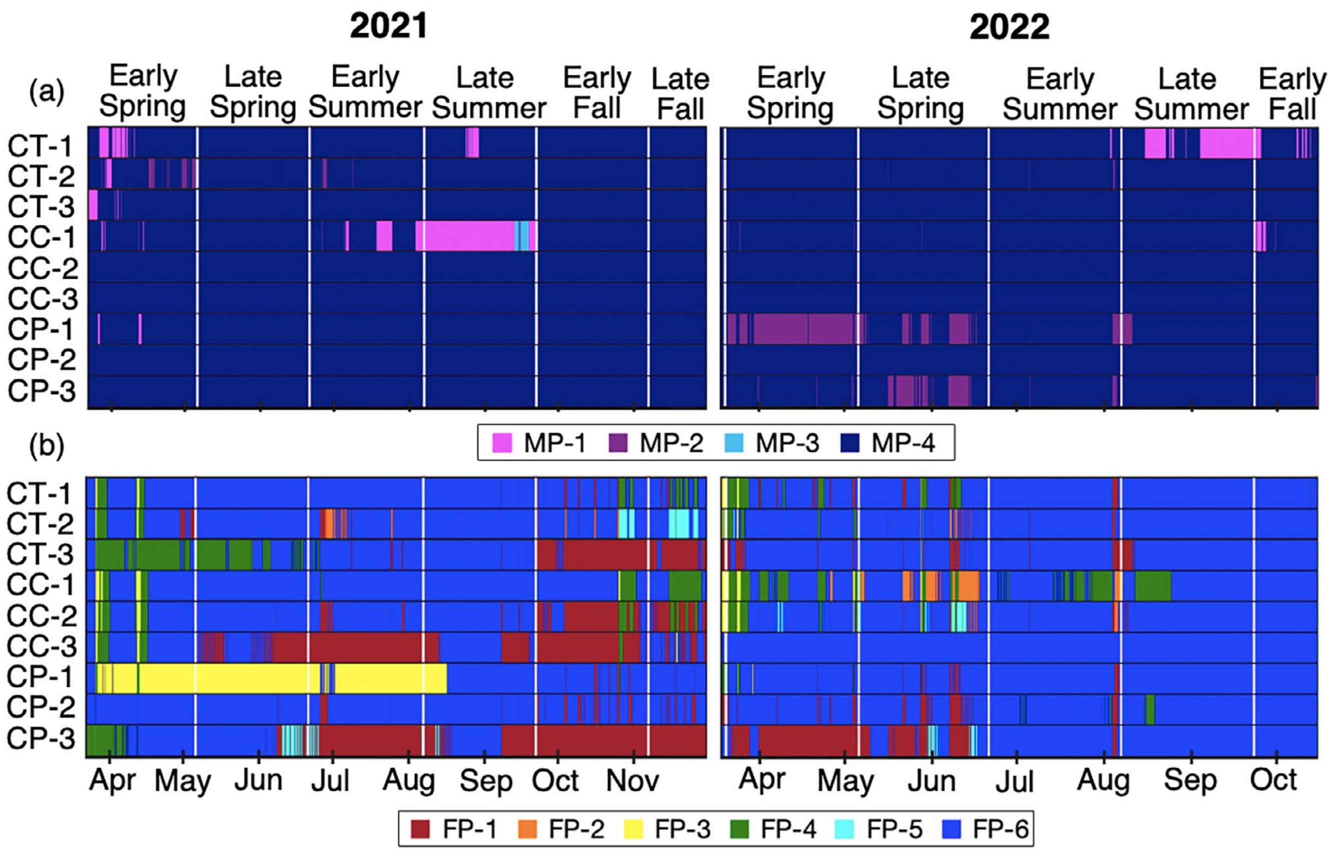

FIGURE 4 | Panel (a) is a categorical timeseries representing the occurrence of different soil moisture patterns of variability (MP): MP-1 (soil moisture increases with depth), MP-2 (soil moisture decreases with depth), MP-3 (soil moisture is uniform) and MP-4 (soil moisture is non-monotonic). Panel (b) is a categorical timeseries representing the occurrence of different dominant flow pathways (FP): FP-1 (wetting front propagation), FP-2 and FP-3 (lateral preferential flow at 30 and 40 cm, respectively), FP-4 (capillary rise), FP-5 (simultaneous wetting front and capillary rise) and FP-6 (no dominant flow pathway). Vertical white lines delimitate the main seasons of interest, while horizontal black lines separate plots. Plot names are shown along the *y*-axis. Vertical white lines delineate the main seasons of interest, while horizontal black lines separate plots. Plot names are shown along the *y*-axis.

TABLE 3 | Spearman's rank correlation coefficients (p-value < 0.05), calculated to compare inter-treatment total soil water storage results.

	Spearman rank correlation coefficients	
Treatments	**2021**	**2022**
CT/CC	0.91	0.96
CT/CP	0.92	0.92
CC/CP	0.85	0.96

which is common for spring in Southern Ontario (D. M. Brown et al. 2013). The decline in soil water storage observed in August 2021 was due to drier conditions (i.e., high evapotranspiration, low precipitation) paired with crop water uptake. During the fall 2021 season, cooler temperatures and longer-duration rainfall events are also typical in Southern Ontario (Pawlick et al. 2019): these weather patterns explain the general increase in total soil water depth observed at the experimental plots during that season. In 2022, recharge was not captured in late summer into early fall, due to it being a considerably drier year than usual—the sixth driest year for the Waterloo Region since their weather records began in 1914 (Seglenieks 2022). These drier conditions also resulted in less soil water storage fluctuations in the latter half of the 2022 growing season when compared to the same period in 2021. It is worth noting that in the present study, the total soil water depth illustrates aggregated soil water storage across the root zone (0–60 cm): while the total storage values may appear stable—or only weakly to moderately variable—during some portions of the growing season, it does not mean that there was no redistribution of water between different soil depths within the root zone.

Comparisons of total soil water depth within treatments, using Spearman's rank correlation coefficients, suggested that the cover crop replicates behaved the most differently from one another. This variability may be due to an unevenly planted cover crop or due to micro-topographical differences of the replicates within the cover crop treatment. Heterogeneity between replicates demonstrates the importance of root systems or networks especially in preferential flow generation, warranting further investigation into the significance of these systems at directing water. While strong inter-treatment Spearman's rank correlations were observed (i.e., treatments exhibited similar 'average' changes in soil water storage), caution is advised when interpreting these results, as this analysis

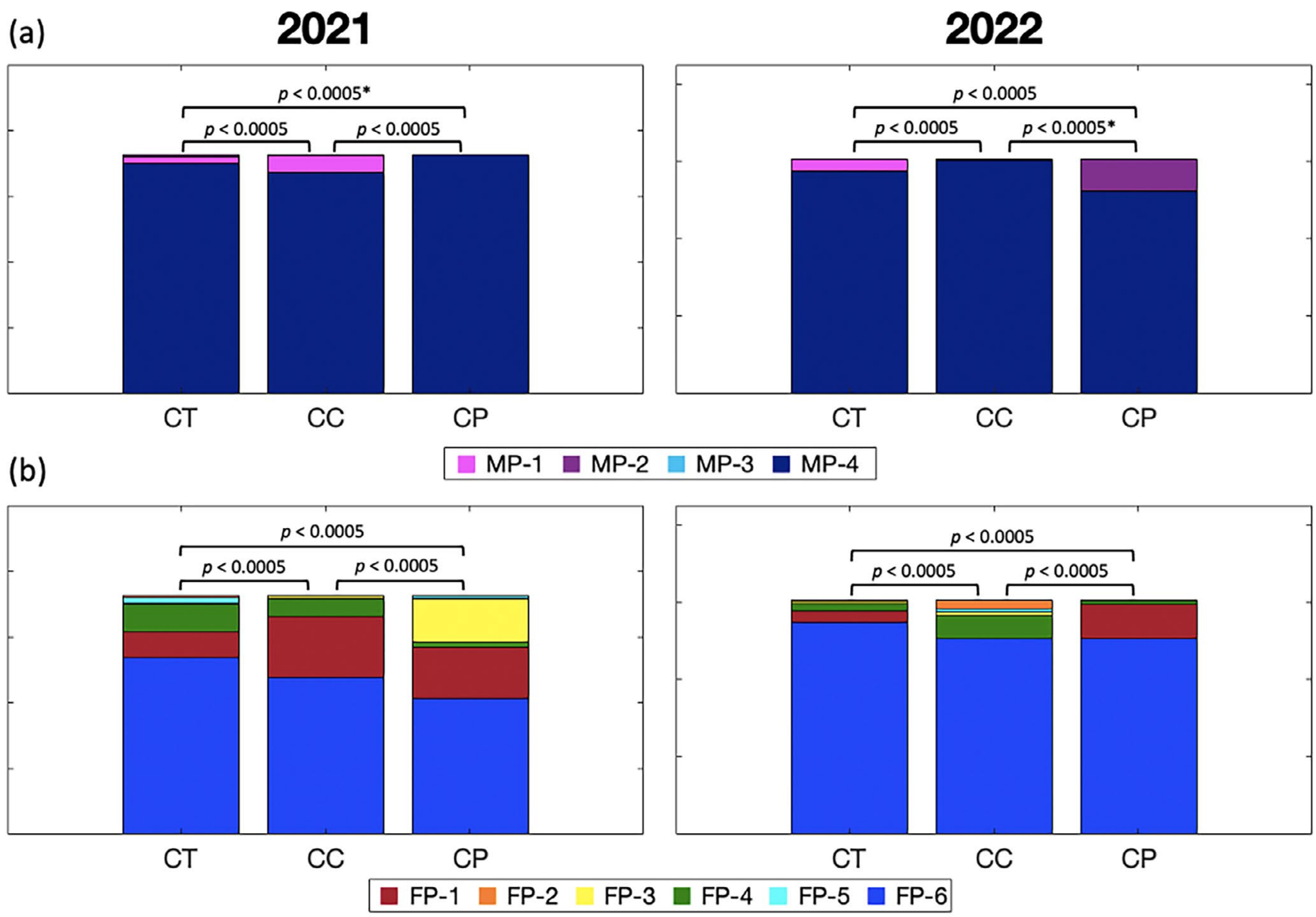

FIGURE 5 | Stacked bar charts illustrating for each treatment the relative occurrence of each pattern of variability class (MP; panel (a)) and each flow pathway class (FP; panel (b)). MPs include: MP-1 (soil moisture increases with depth), MP-2 (soil moisture decreases with depth), MP-3 (soil moisture is uniform) and MP-4 (soil moisture is non-monotonic). FPs include: FP-1 (wetting front propagation), FP-2 and FP-3 (lateral preferential flow at 30 and 40 cm, respectively), FP-4 (capillary rise), FP-5 (simultaneous wetting front and capillary rise) and FP-6 (no dominant flow pathway). Pearson's chi-square test *p*-values calculated between each treatment pair are displayed between the associated bars. The asterisks (*) indicate when a Fisher Exact Test was conducted.

does not assess the magnitude of changes, which descriptive statistics can provide. In the present study, the cover crop (CC) plots had the highest coefficients of variation, meaning that their total soil water depth values varied the most around the mean. This observation was likely in part due to additional evapotranspiration by the cover crop (e.g., Daryanto et al. 2018), especially later into the growing season once the cover crop was established. Throughout the study period, cover crop and control plots had the highest soil water storage fluctuations (i.e., amplitude), likely through better water infiltration compared to the compacted treatment. Several studies conducted at silt-loam sites across the United States have previously reported increases in water infiltration with the use of cover crops and no-till (Adeli et al. 2020; Nouri et al. 2019; Steele et al. 2012). Therefore, increased infiltration capacity and additional evapotranspiration likely contributed to larger soil water storage fluctuations for the cover crop treatment.

When relating water storage to cover crop growth period, increased evapotranspiration impacts on soil water was noticeable in the late summer/early fall of 2021; as the control plots continued to have the highest average soil water storage. All plots had a winter cover crop (planted late August 2020) prior to corn planting, hence some impacts from this cover crop (i.e., plant root and earthworm-generated macropores) may still have been present on the control plots, but they were unlikely to still be intact following the compaction treatment. Although evapotranspiration was not measured directly on the plots, cover crop biomass was measured as part of another multi-year study at the same field site. As expected, less biomass was generated from the cover crop intersown with corn (~0.2 Mg biomass ha^{-1}), compared to the cover crop planted on its own after winter wheat (~3 Mg biomass ha^{-1}) (Tariq et al. 2024), suggesting that the control and cover crop treatments may not have had substantial differences in soil condition. Conversely, for the compacted treatment (CP), the maximum total soil water storage values were the smallest of all treatments, suggesting that water storage capacity may be restricted by compacted soil conditions due to increased bulk density and decreased soil porosity (Da Silva et al. 2008). This is consistent with findings from Nagy et al. (2018), who reported that a compacted plot had 4.2 times less soil water storage than a non-compacted plot in sandy-loam soil. Low soil water storage is a great concern for crops, especially under drought conditions when precipitation is limited. Our results indicate that the cover crop treatment and the compacted treatment differed the most, with respect to soil water storage, thereby suggesting that differences in soil conditions and plant water use between those two treatments were responsible for the observed differences in water storage behaviour.

4.2 | Variations in Inferred Matrix Flow: Seasonal and Soil Condition Influences

Towards the beginning of the early spring in 2021, five of the nine plots had instances of soil moisture increasing monotonically with depth, suggesting that the water table may have still been relatively high following the winter season, thereby contributing to the higher volumetric soil moisture at the bottom of the root zone (60 cm). Near the end of the early spring, some plots had instances of soil moisture decreasing monotonically with depth, which coincided with spring precipitation inputs (Figure 3g) generating top-down soil moisture migration. However, in general, soil moisture patterns for all plots fell within the non-monotonic category for both years. Non-monotonic soil moisture profiles have been reported in previous seasonal-scale studies and typically attributed to differences in soil bulk density and porosity at different depths (Rosenbaum et al. 2012). In the drier year, 2022, all treatments had less moisture pattern variability (i.e., a smaller number of classes and a different temporal distribution of said classes, were observed) than in 2021.

The occurrence of different matrix flow pathways varied both within individual seasons but also between seasons. Generally, the late spring and late summer were dominated by the 'no dominant flow pathway' category, consistent with the drier weather conditions during these seasons, resulting in insufficient water input to generate detectable subsurface flow (i.e., soil moisture did not exceed field capacity). A limitation of this 'no dominant flow' category is that it is essentially a 'catch-all', limiting further process inference. The spring and fall seasons were associated with a greater diversity of flow pathways, attributed to the greater water input fluxes and lower output fluxes typically observed in Southern Ontario (D. M. Brown et al. 2013; Plach et al. 2019). Increases in precipitation and decreases in evapotranspiration during those seasons likely led to more instances of wetting front propagation and capillary rise.

Inferring the presence of capillary rise during these periods would suggest that bottom-up groundwater rise is occurring above the standard tile drain installation depth, which may indicate periods of increased tile flow (Vidon and Cuadra 2010). While it is known that tile drains can impact water dynamics, at the time of sensor installation, the position of the plots and sensors with respect to the tile drains was unknown. The tile drains were installed over 50 years ago and no detailed maps of them are available, although it is unlikely that fields would have been systemically tiled back then. A study by McCoy et al. (2005) conducted at the same research site reported that under 'normal' precipitation years, the water table rises to depths within 1 m of the soil surface during the non-growing season. They also explained that in humid regions like Southern Ontario, soil moisture dynamics are often directly related to the local water table depth. The capillary rise class was four times more common for the control and cover crop treatments compared to the compacted treatment, which is contrary to the findings of at least one published study. Indeed, Huo et al. (2021) evaluated how an increase in the degree of soil compaction can influence capillary rise in silt soils. They found that with increased compaction, soil becomes denser (i.e., particles packed closer together), decreasing pore diameter, resulting in higher capillary water rise. However, this relationship is limited by higher moisture content, as water already contained in soil pores will weaken the capillary effect (Huo et al. 2021). Over the course of the study period, the compacted treatment had twice as much wetting front propagation compared to the other treatments. This observation is consistent with other studies that have found water flow to be restricted due to increased bulk density associated with soil compaction (Keller et al. 2019; Menon et al. 2015), which could potentially limit the amount of downward water movement.

4.3 | Variations in Inferred Preferential Flow: Seasonal and Soil Condition Influences

The overall occurrence of lateral preferential flow was minimal (4% of all timesteps), except for one of the compacted plots (CP-1; Figure 5). However, lateral preferential flow was more common at 40 cm depth rather than the 30 cm depth, suggesting that hydraulic conditions at that depth may promote lateral rather than vertical flow. At the study site, the interface between the Ap and Bt horizons is located at ~32 cm depth (Table 1), which is above the soil moisture sensor located at 40 cm: the Ap-Bt horizon interface therefore does not correspond to a textural discontinuity that could explain the inferred lateral preferential flow at 40 cm in 2021. A more plausible hypothesis is that lateral preferential flow occurs along a compaction-induced plough pan, which is typically located just beneath cultivation depth (~10–25 cm) and usually 5–8 cm thick (Grant et al. 2019; Hayes et al. 2020). Due to the previous decades of agricultural management on this field, a plough pan is likely present across all nine plots but especially so on the compacted (CP) plots to which extra compaction was applied in 2020 and 2021. Lateral preferential flow at 30 cm (1% of all observations) was more common in 2022, potentially driven by shallower movement along the Ap-Bt horizon interface and it occurred most frequently on one cover crop plot (CC-1; Figure 4b).

The fact that 2022 was considerably drier than normal (or compared to 2021) likely contributed to the general lack of active lateral preferential flow, as well as the lack of variability in flow pathways observed in 2022. Soil type can be a driver of preferential flow susceptibility; fine-textured clay soils are more prone to desiccation cracks under dry antecedent conditions which create pathways for preferential flow (Hardie et al. 2011; Nimmo 2012). Although matrix flow is more common in loam soil, higher preferential flow activation has been reported with increased antecedent wetness conditions (Kung et al. 2000). However, Grant et al. (2019) found preferential flow in silt loams to be less affected by antecedent moisture. Similarly, Pluer et al. (2020) found no significant difference in the percent of preferential flow when comparing the growing and the non-growing season. Although data from additional years with average or higher than average total precipitation were not encompassed in this study, in wetter (but still unsaturated) conditions we would expect to observe more variability in moisture patterns and dominant subsurface flow pathways with the availability of more water.

This study used flow pattern analysis to infer the presence of lateral preferential flow using a single multi-depth sensor on each plot. A review of methodologies for detecting preferential flow by Nimmo et al. (2025) suggested that additional sensors would be necessary to confirm the occurrence of lateral preferential flow in each location. While multiple sensors within a plot would be beneficial for verification, the added installations would increase potential disturbance to the soil structure and make crop planting increasingly difficult. Our results showed statistically significant within and inter-treatment differences with respect to the occurrence of moisture pattern classes and matrix flow pathways. However, the true hydrological importance of these differences may not be as significant, as the majority of both years was characterized by the non-monotonic profiles (MP-4) with no dominant flow pathway (FP-6; Figure 5). That said, of the processes identified in these two drier-than-normal years, inter-treatment variability was stronger than within-treatment variability.

5 | Conclusion

In this study, seasonal soil moisture behaviour was compared under different agricultural soil conditions, during two drier-than-normal years, with a special focus on total soil water depth and inferences of subsurface flow. Overall, the findings suggest that the interpretation of results depends on the characteristic measured. For instance, Spearman's rank correlation coefficients revealed similar total soil water storage behaviour for all treatments, but treatment differences were identified as statistically significant when comparing the occurrence of different MP classes and FP classes (Figure 5). With that being said, most of the study period was characterized by a non-monotonic soil moisture profile with no dominant flow pathway. When dominant flow pathways were active, results from within-treatment (replicate-to-replicate) tests demonstrated spatial heterogeneity, while subtle differences in inter-treatment dynamics were also identified. These observations highlight the importance of not overlooking detailed soil moisture profile analyses, as they capture depth-to-depth variations caused by soil water re-distribution, when studying soil water storage and flow processes. Although the findings of this study are particular to the region of our experimental site (Southern Ontario), advancements in soil moisture sensing technology and increased availability of sensors allow this type of data collection and analysis to be conducted feasibly in other regions with varying climate, soil and ecological conditions. The approach used here provides additional tools to interpret soil moisture time series under different management that could be used to assess when preferential flow is likely to occur. Determining which soil conditions are more susceptible to preferential flow is useful for predicting future locations where rapid soil water movement, potentially carrying contaminants to tile drains and groundwater, can be expected.

Acknowledgements

This research was supported through funding provided by the Ontario Ministry of Agriculture, Food and Agribusiness (OMAFA), through the Ontario Agri-Food Innovation Alliance. We gratefully acknowledge Sean Jordan for his tremendous support with experimental conceptualization and field assistance and we thank Dr. Shannon Brown for her expertise and help with data collection. We extend our gratitude to Jack Moore, Anika Chiang, Noelle Starling, Nicholas Puopolo, Alexander Figliuzzi, Amina Hassan and Jamie Bain for their assistance in the field.

Funding

This work was supported by Ontario Ministry of Agriculture, Food and Agribusiness.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

Adeli, A., J. P. Brooks, J. J. Read, et al. 2020. "Management Strategies on an Upland Soil for Improving Soil Properties." *Communications in Soil Science and Plant Analysis* 51, no. 3: 413–429. https://doi.org/10.1080/00103624.2019.1709490.

Ali, G., M. Macrae, M. Walker, J. Laing, and D. Lobb. 2018. "Preferential Flow in Vertisolic Soils With and Without Organic Amendments." *Agricultural & Environmental Letters* 3, no. 1: 1–5. https://doi.org/10.2134/ael2018.04.0018.

Allaire, S. E., S. Roulier, and A. J. Cessna. 2009. "Quantifying Preferential Flow in Soils: A Review of Different Techniques." *Journal of Hydrology* 378, no. 1–2: 179–204. https://doi.org/10.1016/j.jhydrol.2009.08.013.

Alonso-Ayuso, M., J. L. Gabriel, and M. Quemada. 2014. "The Kill Date as a Management Tool for Cover Cropping Success." *PLoS One* 9, no. 10: e109587. https://doi.org/10.1371/journal.pone.0109587.

Arora, B., D. Dwivedi, B. Faybishenko, R. B. Jana, and H. M. Wainwright. 2019. "Understanding and Predicting Vadose Zone Processes." *Reviews in Mineralogy and Geochemistry* 85, no. 1: 303–328. https://doi.org/10.2138/rmg.2019.85.10.

Basche, A., and M. DeLonge. 2017. "The Impact of Continuous Living Cover on Soil Hydrologic Properties: A Meta-Analysis." *Soil Science Society of America Journal* 81, no. 5: 1179–1190. https://doi.org/10.2136/sssaj2017.03.0077.

Basche, A. D., and M. S. DeLonge. 2019. "Comparing Infiltration Rates in Soils Managed With Conventional and Alternative Farming Methods: A Meta-Analysis." *PLoS One* 14, no. 9: 1–22. https://doi.org/10.1371/journal.pone.0215702.

Bauke, S. L., W. Amelung, R. Bol, et al. 2022. "Soil Water Status Shapes Nutrient Cycling in Agroecosystems From Micrometer to Landscape Scales." *Journal of Plant Nutrition and Soil Science* 185, no. 6: 773–792. https://doi.org/10.1002/jpln.202200357.

Beven, K., and P. Germann. 1982. "Macropores and Water Flow in Soils." *Water Resources Research* 18, no. 5: 1311–1325. https://doi.org/10.1029/WR018i005p01311.

Blanchy, G., C. W. Watts, J. Richards, et al. 2020. "Time-Lapse Geophysical Assessment of Agricultural Practices on Soil Moisture Dynamics." *Vadose Zone Journal* 19, no. 1: 1–15. https://doi.org/10.1002/vzj2.20080.

Blanco-Canqui, H., M. M. Mikha, D. R. Presley, and M. M. Claassen. 2011. "Addition of Cover Crops Enhances no-Till Potential for Improving Soil Physical Properties." *Soil Science Society of America Journal* 75, no. 4: 1471–1482. https://doi.org/10.2136/sssaj2010.0430.

Blanco-Canqui, H., and S. J. Ruis. 2020. "Cover Crop Impacts on Soil Physical Properties: A Review." *Soil Science Society of America Journal* 84, no. 5: 1527–1576. https://doi.org/10.1002/saj2.20129.

Brown, D. M., H. Dadfar, D. J. Fallow, R. J. Gordon, J. D. Lauzon, and G. W. Parkin. 2013. "Temporal and Spatial Variability of Water Surplus in Ontario, Canada." *ISRN Soil Science* 2013: 1–7. https://doi.org/10.1155/2013/362895.

Brown, S., C. Wagner-Riddle, Z. Debruyn, et al. 2021. "Assessing Variability of Soil Water Balance Components Measured at a New Lysimeter Facility Dedicated to the Study of Soil Ecosystem Services." *Journal of Hydrology* 603: 127037. https://doi.org/10.1016/j.jhydrol.2021.127037.

Chalise, K. S., S. Singh, B. R. Wegner, et al. 2018. "Cover Crops and Returning Residue Impact on Soil Organic Carbon, Bulk Density, Penetration Resistance, Water Retention, Infiltration, and Soybean Yield." *Agronomy Journal* 111, no. 1: 99–108. https://doi.org/10.2134/agronj2018.03.0213.

Clothier, B. E., S. R. Green, and M. Deurer. 2008. "Preferential Flow and Transport in Soil: Progress and Prognosis." *European Journal of Soil Science* 59, no. 1: 2–13. https://doi.org/10.1111/j.1365-2389.2007.00991.x.

Colombi, T., L. C. Torres, A. Walter, and T. Keller. 2018. "Feedbacks Between Soil Penetration Resistance, Root Architecture and Water Uptake Limit Water Accessibility and Crop Growth – A Vicious Circle." *Science of the Total Environment* 626: 1026–1035. https://doi.org/10.1016/j.scitotenv.2018.01.129.

Da Silva, S. R., N. F. De Barros, L. M. Da Costa, and F. P. Leite. 2008. "Soil Compaction and Eucalyptus Growth in Response to Forwarder Traffic Intensity and Load." *Revista Brasileira de Ciência do Solo* 32, no. 3: 921–932. https://doi.org/10.1590/s0100-06832008000300002.

Daryanto, S., B. Fu, L. Wang, P. A. Jacinthe, and W. Zhao. 2018. "Quantitative Synthesis on the Ecosystem Services of Cover Crops." *Earth-Science Reviews* 185: 357–373. https://doi.org/10.1016/j.earscirev.2018.06.013.

Dunne, T., T. R. Moore, and C. H. Taylor. 1975. "Recognition and Prediction of Runoff-Producing Zones in Humid Regions." *Hydrological Sciences Bulletin des Sciences Hydrologiques* 20, no. 3: 305–327.

Environment Canada. 2023. *Historical Climate Data*. Elora RCS.

Frieman, J., D. A. Saucier, and S. S. Miller. 2017. "Testing Hypotheses When the Dependent Variable Consists of Frequencies of Scores in Various Categories." In *Principles & Methods of Statistical Analysis*, 1st ed., 214–241. SAGE Publications, Inc. https://doi.org/10.4135/9781071878903.

Gao, Y., K. A. Borden, S. E. Brown, and C. Wagner-Riddle. 2023. "Nongrowing Season Soil Nitrous Oxide Emissions as Influenced by Cover Crops and Fall Tillage Termination." *Canadian Journal of Soil Science* 1–11: 527–537. https://doi.org/10.1139/cjss-2023-0017.

Gjettermann, B., K. L. Nielsen, C. T. Petersen, H. E. Jensen, and S. Hansen. 1997. "Preferential Flow in Sandy Loam Soils as Affected by Irrigation Intensity." *Soil Technology* 11, no. 2: 139–152. https://doi.org/10.1016/S0933-3630(97)00001-9.

Graham, C. B., and H. S. Lin. 2012. "The Hydropedograph Toolbox and Its Application." *Hydrology and Earth System Sciences Discussions* 9, no. 12: 14231–14271. https://doi.org/10.5194/hessd-9-14231-2012.

Grant, K. N., M. L. Macrae, and G. A. Ali. 2019. "Differences in Preferential Flow With Antecedent Moisture Conditions and Soil Texture: Implications for Subsurface P Transport." *Hydrological Processes* 33: 2068–2079. https://doi.org/10.1002/hyp.13454.

Hardie, M. A., W. E. Cotching, R. B. Doyle, G. Holz, S. Lisson, and K. Mattern. 2011. "Effect of Antecedent Soil Moisture on Preferential Flow in a Texture-Contrast Soil." *Journal of Hydrology* 398, no. 3–4: 191–201. https://doi.org/10.1016/j.jhydrol.2010.12.008.

Haruna, S. I., E. Ritchey, C. Mosley, and S. Ku. 2022. "Effects of Cover Crops on Soil Hydraulic Properties During Commodity Crop Growing Season." *Soil Use and Management* 39, no. 1: 218–231. https://doi.org/10.1111/sum.12803.

Hayes, A., D. Aspinall, A. Barrie, et al. 2020. *Best Management Practices Subsurface Compaction*. Queen's Printer for Ontario. https://bmpbooks.com/media/Subsurface-Compaction.pdf.

Huo, W., Z. Zhu, Y. Peng, S. Pu, Y. Wan, and C. Zhang. 2021. "Experimental Study on Capillary Water Rise and Influencing Factors in Silty Soil." *Arabian Journal of Geosciences* 14, no. 24: 2706. https://doi.org/10.1007/s12517-021-09097-6.

IUSS Working Group W.R.B. 2006. "World Reference Base for Soil Resources (WRB)." In *World Soil Resources Rep. 103*. Elsevier. https://doi.org/10.1016/B978-0-12-822974-3.00161-0.

Kampf, S. K. 2011. "Variability and Persistence of Hillslope Initial Conditions: A Continuous Perspective on Subsurface Flow Response to Rain Events." *Journal of Hydrology* 404, no. 3–4: 176–185. https://doi.org/10.1016/j.jhydrol.2011.04.028.

Keller, T., M. Sandin, T. Colombi, R. Horn, and D. Or. 2019. "Historical Increase in Agricultural Machinery Weights Enhanced Soil Stress Levels and Adversely Affected Soil Functioning." *Soil and Tillage Research* 194: 104293. https://doi.org/10.1016/j.still.2019.104293.

Kottek, M., J. Grieser, C. Beck, B. Rudolf, and F. Rubel. 2006. "World Map of the Köppen-Geiger Climate Classification Updated." *Meteorologische Zeitschrift* 15, no. 3: 259–263. https://doi.org/10.1127/0941-2948/2006/0130.

Kung, K.-J. S., T. S. Steenhuis, E. J. Kladivko, T. J. Gish, G. Bubenzer, and C. S. Helling. 2000. "Impact of Preferential Flow on the Transport of Adsorbing and Non-Adsorbing Tracers." *Soil Science Society of America Journal* 64, no. August: 1050–1052. https://doi.org/10.2136/sssaj2000.6441290x.

Lai, X., Z. Zhou, Q. Zhu, and K. Liao. 2018. "Comparing the Spatio-Temporal Variations of Soil Water Content and Soil Free Water Content at the Hillslope Scale." *Catena* 160: 366–375. https://doi.org/10.1016/j.catena.2017.10.008.

Lewan, E., J. Kreuger, and N. Jarvis. 2009. "Implications of Precipitation Patterns and Antecedent Soil Water Content for Leaching of Pesticides From Arable Land." *Agricultural Water Management* 96, no. 11: 1633–1640. https://doi.org/10.1016/j.agwat.2009.06.006.

Lin, H., and X. Zhou. 2008. "Evidence of Subsurface Preferential Flow Using Soil Hydrologic Monitoring in the Shale Hills Catchment." *European Journal of Soil Science* 59, no. 1: 34–49. https://doi.org/10.1111/j.1365-2389.2007.00988.x.

Lu, N., and W. J. Likos. 2004. "Rate of Capillary Rise in Soil." *Journal of Geotechnical and Geoenvironmental Engineering* 130, no. 6: 646–650. https://doi.org/10.1061/(asce)1090-0241(2004)130:6(646).

Machado, P. V. F., R. E. Farrell, W. Deen, R. P. Voroney, K. A. Congreves, and C. Wagner-Riddle. 2021. "Contribution of Crop Residue, Soil, and Fertilizer Nitrogen to Nitrous Oxide Emissions Varies With Long-Term Crop Rotation and Tillage." *Science of the Total Environment* 767: 145107. https://doi.org/10.1016/j.scitotenv.2021.145107.

McCoy, A. J., G. Parkin, C. Wagner-Riddle, et al. 2005. "Using Automated Soil Water Content Measurements to Estimate Soil Water Budgets." *Canadian Journal of Soil Science* 86, no. 1: 47–56. https://doi.org/10.4141/S05-031.

Menheere, N. 2023. "Mitigating Nitrous Oxide Emissions Using Nitrification and Urease Inhibitors in Corn-Based Rotations Following Cover Crops Termination." Master's thesis. University of Guelph.

Menon, M., X. Jia, G. J. Lair, P. H. Faraj, and A. Blaud. 2015. "Soil & Tillage Research Analysing the Impact of Compaction of Soil Aggregates Using X-Ray Microtomography and Water Flow Simulations." *Soil & Tillage Research* 150: 147–157. https://doi.org/10.1016/j.still.2015.02.004.

Moragoda, N., M. Kumar, and S. Cohen. 2022. "Representing the Role of Soil Moisture on Erosion Resistance in Sediment Models: Challenges and Opportunities." *Earth-Science Reviews* 229: 104032. https://doi.org/10.1016/j.earscirev.2022.104032.

Nagy, V., P. Šurda, Ľ. Lichner, A. J. Kovács, and G. Milics. 2018. "Impact of Soil Compaction on Water Content in Sandy Loam Soil Under Sunflower." *Journal of Hydrology and Hydromechanics* 66, no. 4: 416–420. https://doi.org/10.2478/johh-2018-0036.

Nawaz, M. F., G. Bourrié, and F. Trolard. 2013. "Soil Compaction Impact and Modelling. A Review." *Agronomy for Sustainable Development* 33, no. 2: 291–309. https://doi.org/10.1007/s13593-011-0071-8.

Nimmo, J. R. 2012. "Preferential Flow Occurs in Unsaturated Conditions." *Hydrological Processes* 26, no. 5: 786–789. https://doi.org/10.1002/hyp.8380.

Nimmo, J. R., I. Wiekenkamp, R. Araki, et al. 2025. "Identifying Preferential Flow From Soil Moisture Time Series: Review of Methodologies." *Vadose Zone Journal* 24, no. 2: 1–25. https://doi.org/10.1002/vzj2.70017.

Nouri, A., J. Lee, X. Yin, D. D. Tyler, and A. M. Saxton. 2019. "Thirty-Four Years of No-Tillage and Cover Crops Improve Soil Quality and Increase Cotton Yield in Alfisols, Southeastern USA." *Geoderma* 337, no. November 2018: 998–1008. https://doi.org/10.1016/j.geoderma.2018.10.016.

Ogilvie, C. M., W. Ashiq, H. B. Vasava, and A. Biswas. 2021. "Quantifying Root-Soil Interactions in Cover Crop Systems: A Review." *Agriculture (Switzerland)* 11: 218. https://doi.org/10.3390/agriculture11030218.

OMAFA (Ontario Ministry of Agriculture, Food and Agribusiness). 2022. "Climate Zones and Planting Dates for Vegetables in Ontario." https://www.ontario.ca/page/climate-zones-and-planting-dates-vegetables-ontario.

Pawlick, A. A., C. Wagner-Riddle, G. W. Parkin, and A. A. Berg. 2019. "Assessment of Nitrification and Urease Inhibitors on Nitrate Leaching in Corn (*Zea mays* L.)." *Canadian Journal of Soil Science* 99, no. 1: 80–91. https://doi.org/10.1139/cjss-2018-0110.

Peterson, A. M., W. H. Helgason, and A. M. Ireson. 2019. "How Spatial Patterns of Soil Moisture Dynamics Can Explain Field-Scale Soil Moisture Variability: Observations From a Sodic Landscape." *Water Resources Research* 55, no. 5: 4410–4426. https://doi.org/10.1029/2018WR023329.

Pituello, C., N. Dal Ferro, G. Simonetti, A. Berti, and F. Morari. 2016. "Nano to Macro Pore Structure Changes Induced by Long-Term Residue Management in Three Different Soils." *Agriculture, Ecosystems and Environment* 217: 49–58. https://doi.org/10.1016/j.agee.2015.10.029.

Plach, J., W. Pluer, M. Macrae, et al. 2019. "Agricultural Edge-Of-Field Phosphorus Losses in Ontario, Canada: Importance of the Nongrowing Season in Cold Regions." *Journal of Environmental Quality* 48: 813–821. https://doi.org/10.2134/jeq2018.11.0418.

Pluer, W. T., M. Macrae, A. Buckley, and K. Reid. 2020. "Contribution of Preferential Flow to Tile Drainage Varies Spatially and Temporally." *Vadose Zone Journal* 19: e20043. https://doi.org/10.1002/vzj2.20043.

Ranathunga, K. N., J. Evans, N. Toth, S. Brown, L. L. Van Eerd, and C. Wagner-riddle. 2025. "Net Ecosystem Carbon Budget and Net Greenhouse Gas Emissions Under Diverse Crop Rotation Using Cover Crops Compared to a Conventional Crop Rotation." *Agriculture, Ecosystems and Environment* 381: 109418. https://doi.org/10.1016/j.agee.2024.109418.

Rieke, E. L., D. K. Bagnall, C. L. S. Morgan, et al. 2022. "Evaluation of Aggregate Stability Methods for Soil Health." *Geoderma* 428: 116156. https://doi.org/10.1016/j.geoderma.2022.116156.

Rosegrant, M. W., C. Ringler, and T. Zhu. 2009. "Water for Agriculture: Maintaining Food Security Under Growing Scarcity." *Annual Review of Environment and Resources* 34: 205–222. https://doi.org/10.1146/annurev.environ.030308.090351.

Rosenbaum, U., H. R. Bogena, M. Herbst, et al. 2012. "Seasonal and Event Dynamics of Spatial Soil Moisture Patterns at the Small Catchment Scale." *Water Resources Research* 48, no. 10: 1–22. https://doi.org/10.1029/2011WR011518.

Ruosteenoja, K., T. Markkanen, A. Venäläinen, P. Räisänen, and H. Peltola. 2018. "Seasonal Soil Moisture and Drought Occurrence in Europe in CMIP5 Projections for the 21st Century." *Climate Dynamics* 50, no. 3–4: 1177–1192. https://doi.org/10.1007/s00382-017-3671-4.

Saso, J. K., G. W. Parkin, C. F. Drury, J. D. Lauzon, and W. D. Reynolds. 2012. "Chloride Leaching in Two Ontario Soils: Measurement and Prediction Using HYDRUS-1D." *Canadian Journal of Soil Science* 92, no. 2: 285–296. https://doi.org/10.4141/cjss2011-046.

Seglenieks, F. 2022. University of Waterloo Weather Station Annual Summary - 2022. Data Archives - Eric D. Soulis Weather Station. https://weather.uwaterloo.ca/download/historical/2022/UW_weather_station_summary_2022.pdf.

Seneviratne, S. I., T. Corti, E. L. Davin, et al. 2010. "Investigating Soil Moisture-Climate Interactions in a Changing Climate: A Review." *Earth-Science Reviews* 99, no. 3–4: 125–161. https://doi.org/10.1016/j.earscirev.2010.02.004.

Shein, E. V., D. I. Shcheglov, A. B. Umarova, I. V. Sokolova, and E. Y. Milanovskii. 2009. "Structural Status of Technogenic Soils and the Development of Preferential Water Flows." *Eurasian Soil Science* 42, no. 6: 636–644. https://doi.org/10.1134/S1064229309060088.

Soil Classification Working Group. 1998. "The Canadian System of Soil Classification (No. 1646; 3rd ed.)".

Steele, M. K., F. J. Coale, and R. L. Hill. 2012. "Winter Annual Cover Crop Impacts on no-Till Soil Physical Properties and Organic Matter." *Soil Science Society of America Journal* 76, no. 6: 2164–2173. https://doi.org/10.2136/sssaj2012.0008.

Stumpp, C., and G. Kammerer. 2022. "The Vadose Zone—A Semi-Aquatic Ecosystem." In *Encyclopedia of Inland Waters*, edited by D. Bouffard, M.-E. Perga, and L. Poff, 2nd ed., 331–338. Elsevier Inc.

Tariq, A., N. Menheere, Y. Gao, et al. 2024. "Increased N2O Emissions by Cover Crops in a Diverse Crop Rotation Can Be Mediated With Dual Nitrification and Urease Inhibitors." *Agriculture, Ecosystems and Environment* 374, no. July: 109178. https://doi.org/10.1016/j.agee.2024.109178.

Tribouillois, H., J. Constantin, L. Casal, J. Villerd, and O. Therond. 2022. "Introducing and Expanding Cover Crops at the Watershed Scale: Impact on Water Flows." *Agriculture, Ecosystems and Environment* 337: 108050. https://doi.org/10.1016/j.agee.2022.108050.

Triola, M. M., M. F. Triola, and J. Roy. 2018. *Biostatistics for the Biological and Health Sciences*. 2nd ed. Pearson Education Inc.

Van Eerd, L. L., I. Chahal, Y. Peng, and J. C. Awrey. 2023. "Influence of Cover Crops at the Four Spheres: A Review of Ecosystem Services, Potential Barriers, and Future Directions for North America." *Science of the Total Environment* 858: 159990. https://doi.org/10.1016/j.scitotenv.2022.159990.

Vidon, P., and P. E. Cuadra. 2010. "Impact of Precipitation Characteristics on Soil Hydrology in Tile-Drained Landscapes." *Hydrological Processes* 24, no. 13: 1821–1833. https://doi.org/10.1002/hyp.7627.

Zhang, H., W. Li, Z. Zou, et al. 2025. "Subsurface Lateral Flow in Steep Granite Slope and Its Relation to Benggang Erosion." *Hydrological Processes* 39: 1–15. https://doi.org/10.1002/hyp.70157.

Zhang, Y., Z. Zhang, Z. Ma, et al. 2018. "A Review of Preferential Water Flow in Soil Science." *Canadian Journal of Soil Science* 98, no. 4: 604–618. https://doi.org/10.1139/cjss-2018-0046.

Zhu, Q., X. Nie, X. Zhou, K. Liao, and H. Li. 2014. "Soil Moisture Response to Rainfall at Different Topographic Positions Along a Mixed Land-Use Hillslope." *Catena* 119: 61–70. https://doi.org/10.1016/j.catena.2014.03.010.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Irrigation experiment procedure. **Figure S1:** Continuous one-minute frequency soil moisture (expressed as percent volume) measured during the irrigation experiment on CT-2, with later identified field capacity value. **Table S1:** Field capacity values (%vol) for each plot and each target depth (20, 30, 40 and 60 cm). **Table S2:** Spearman's rank correlation coefficients (p-value < 0.05), calculated to compare total soil water storage (computed for each 15-min. timestep) results among replicates of the same treatment in 2021 ($n = 24266$) and 2022 ($n = 20246$). **Table S3:** Pearson's chi-square test p-values, used to assess the homogeneity of replicates of the same treatment with respect to frequency counts of moisture pattern (MP) classes and flow pathway (FP) classes. N/A indicates cases where two plots had the exact same frequency counts of each MP class or FP class.